

cURL — Practical Tool Guide

HTTP from the terminal: GET, POST, JSON, headers, cookies, authentication, files, and proxying traffic through Burp.

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Usage note: A focused standalone cURL reference for web labs and request troubleshooting.

REQUEST

Choose URL / method

HEADERS

Add auth / cookies

PROXY

Send through Burp

VERIFY

Review the response

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1. Basic syntax

cURL is a fast way to test HTTP directly from the terminal.

KALI TERMINAL — Simple GET

```
curl http://[TARGET]/
```

KALI TERMINAL — Headers only

```
curl -I http://[TARGET]/
```

KALI TERMINAL — Headers + body

```
curl -i http://[TARGET]/
```

KALI TERMINAL — Connection details

```
curl -v http://[TARGET]/
```

Verbose output helps with DNS, TLS, request, and response details, but it can be large.

2. Important options

Option	Meaning	When to use it
-L	Follow redirects	When 301/302 redirects must be followed to the final page.
-X	Choose method	For PUT/DELETE etc.; do not add it unnecessarily when -d already implies POST.
-H	Add a header	Host, Authorization, Content-Type, and custom headers.
-d	POST form data	Usually application/x-www-form-urlencoded.
--json	Send JSON with appropriate headers	Convenient for APIs.
-F	Multipart form/upload	File upload or multipart forms.
-b	Send cookie or read cookie file	Authenticated sessions.
-c	Save cookies to a file	Reuse a session across requests.
-u	Basic authentication	user:pass.
-k	Ignore TLS certificate validation	Only in labs with self-signed certificates.
-x	Proxy	Send traffic through Burp.
-o	Save body to a file	Download a response body.

3. GET / POST / JSON

KALI TERMINAL — Query string

```
curl "http://[TARGET]/search?q=test"
```

KALI TERMINAL — POST form

```
curl -X POST -d "username=[USER]&password=[PASS]" http://[TARGET]/login
```

KALI TERMINAL — URL-encode a value

```
curl --data-urlencode "q=a b&c" http://[TARGET]/search
```

KALI TERMINAL — POST JSON

```
curl --json '{"name":"test","role":"user"}' http://[TARGET]/api
```

KALI TERMINAL — Custom headers

```
curl -H "X-Test: 1" -H "Accept: application/json" http://[TARGET]/
```

4. Authentication and cookies

KALI TERMINAL — Send cookie directly

```
curl -b "session=[VALUE]" http://[TARGET]/profile
```

KALI TERMINAL — Save and reuse cookies

```
curl -c cookies.txt http://[TARGET]/login  
curl -b cookies.txt http://[TARGET]/profile
```

KALI TERMINAL — Basic Auth

```
curl -u [USER]:[PASS] http://[TARGET]/protected
```

5. TLS, redirects, and files

KALI TERMINAL — Follow redirect and save file

```
curl -L -o output.bin http://[TARGET]/download
```

KALI TERMINAL — Multipart upload

```
curl -F "file=@test.txt" -F "submit=1" http://[TARGET]/upload
```

KALI TERMINAL — Self-signed TLS in a lab

```
curl -k https://[TARGET]/
```

Note: `-k/--insecure` disables certificate validation. Use it only in a lab; it is not a fix for TLS problems in a real environment.

6. Use cURL with Burp

Proxying through Burp is the clearest way to see exactly what cURL sends.

KALI TERMINAL — Burp proxy

```
curl -x http://127.0.0.1:8080 http://[TARGET]/
```

KALI TERMINAL — HTTPS through Burp with an untrusted lab certificate

```
curl -x http://127.0.0.1:8080 -k https://[TARGET]/
```

Run Burp, review Proxy → HTTP history, and combine CLI speed with visible request/response details.

7. Fast troubleshooting

KALI TERMINAL — Retry with verbose output

```
curl -v [URL]
```

KALI TERMINAL — Force HTTP/1.1 when compatibility requires it

```
curl --http1.1 [URL]
```

KALI TERMINAL — Help for a specific option area

```
curl --help all  
curl --help --cookie
```

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

<https://curl.se/docs/manpage.html>